

No. 1 ([H] on $Mo_3Pd(111)$)

- **Iteration 1:** Hollow @ Mo-Mo-Pd → **Slipped** to Pd Ontop (-2.601 eV).
- **Iteration 2:** Bridge @ Mo-Pd → **Slipped** to Pd Ontop (-2.634 eV).
- **Iteration 3:** Ontop @ Pd → **Stable** at Pd Ontop (-2.625 eV).
- **Iteration 4:** Ontop @ Mo → **Slipped (Chemical)** to Pd Ontop (-2.601 eV).
- **Iteration 5:** Bridge @ Mo-Mo → **Slipped** to Pd Ontop (-2.638 eV).
- **Conclusion:** The exploration reveals a highly convergent Potential Energy Surface where almost all initial geometries (including Mo-dominated sites and high-coordination hollow/bridge sites) spontaneously relax to the **Pd ontop site**. This identifies the Pd ontop position as the exclusive thermodynamic sink ($E_{ads} \approx -2.638$ eV) and suggests that Mo sites are kinetically unstable for atomic hydrogen adsorption on this specific surface structure.

No. 2 ([N] = N[H] on $Mo_3Pd(111)$)

- **Iteration 1:** Side-on @ Mo-Pd bridge → **Slipped** to Mo-Mo-Mo Hollow (-3.937 eV).
- **Iteration 2:** End-on (N_{rad}) @ Mo-Mo-Pd hollow → **Slipped** to Mo-Mo bridge (-3.542 eV).
- **Iteration 3:** End-on (N_{rad}) @ Mo top → **Isomerized** to Mo-Mo-Mo Hollow (-3.709 eV).
- **Iteration 4:** Side-on @ Mo-Mo-Mo hollow → **Dissociated** (-3.441 eV).
- **Iteration 5:** Side-on @ Pd-Pd bridge → **Slipped** to Mo-Mo-Mo Hollow (Isomerized, -3.716 eV).
- **Conclusion:** The exploration reveals a funnel-shaped Potential Energy Surface where kinetically unstable Pd sites spontaneously funnel the adsorbate into a deep thermodynamic sink at the **Mo-dominated hollow site**, highlighting a critical trade-off where Mo centers drive strong binding (risking dissociation in specific geometries) while Pd facilitates surface mobility.

No. 3 ([H] on $Pd_3Cu(111)$)

- **Iteration 1:** Hollow @ Pd-Pd-Pd → **Stable** at Pd-Pd-Pd Hollow (-3.000 eV).
- **Iteration 2:** Hollow @ Cu-Pd-Pd → **Slipped** to Pd-Pd-Pd Hollow (-3.018 eV).
- **Iteration 3:** Bridge @ Pd-Pd → **Stable** at Pd-Pd Bridge (-2.929 eV).
- **Iteration 4:** Ontop @ Pd → **Slipped** to Pd-Pd Bridge (-2.965 eV).
- **Iteration 5:** Bridge @ Cu-Pd → **Slipped** to Pd-Pd-Pd Hollow (-3.029 eV).
- **Conclusion:** The exploration demonstrates a strong thermodynamic preference for **pure Pd ensembles**, as the H atom spontaneously slips from all Cu-containing sites (Cu-Pd-Pd hollow, Cu-Pd bridge) to Pd-dominated regions. The Global Minimum is identified at the **Pd-Pd-Pd hollow site** (-3.029 eV), highlighting that Cu acts as a destabilizing agent for H adsorption on this surface, driving the adsorbate toward high-coordination Pd sites.

No. 4 ([N] = N[H] on $Pd_3Cu(111)$)

- **Iteration 1:** Side-on @ Pd-Pd bridge → **Slipped** to Cu-Pd-Pd Hollow (Isomerized, -1.831 eV).
- **Iteration 2:** Side-on @ Cu-Cu-Pd-Pd hollow → **Slipped** to Cu-Cu-Pd-Pd-Pd Hollow (Isomerized, -2.353 eV).
- **Iteration 3:** Side-on @ Cu-Cu-Pd-Pd-Pd hollow → **Dissociated** to Cu Ontop (-3.368 eV).
- **Iteration 4:** Side-on @ Cu-Pd-Pd hollow → **Slipped** to Cu-Cu-Pd-Pd Hollow (Isomerized, -1.842 eV).
- **Iteration 5:** End-on (N_{rad}) @ Cu top → **Dissociated** (-3.442 eV).
- **Conclusion:** The exploration reveals a complex landscape where **mixed Cu-Pd hollow sites** stabilize the molecular form (via isomerization, best molecular $E_{ads} = -2.353$ eV), but the system exhibits a strong thermodynamic drive toward **dissociation** (reaching -3.442 eV), particularly initiated at Cu sites or high-coordination hollows, indicating the radical is metastable and prone to breaking apart on this surface.

No. 5 ($[H]$ on $Cu_3Ag(111)$)

- **Iteration 1:** End-on @ Cu-Cu-Cu Hollow → **Stable** (-2.536 eV).
- **Iteration 2:** End-on @ Ag-Cu-Cu Hollow → **Slipped** to Cu-Cu-Cu Hollow (-2.536 eV).
- **Iteration 3:** End-on @ Cu-Cu Bridge → **Slipped** to Cu-Cu-Cu Hollow (-2.541 eV).
- **Iteration 4:** End-on @ Ag-Cu Bridge → **Slipped** to Cu-Cu-Cu Hollow (-2.558 eV).
- **Iteration 5:** End-on @ Cu Top → **Slipped** to Cu-Cu-Cu Hollow (-2.558 eV).
- **Conclusion:** The exploration reveals a highly convergent Potential Energy Surface where almost all initial sites (including Ag-containing hollows, bridges, and Cu top sites) spontaneously funnel the adsorbate into a deep thermodynamic sink at the **pure Cu-dominated hollow site** (-2.558 eV). This highlights a significant chemical preference for high-coordination Cu sites over Ag sites, with strong "chemical slip" driving the system to the global minimum.

No. 6 ($[N] = N[H]$ on $Cu_3Ag(111)$)

- **Iteration 1:** Side-on @ Cu-Cu Bridge → **Slipped** to Cu-Cu-Cu Hollow (Isomerized) (-1.736 eV).
- **Iteration 2:** Side-on @ Ag-Cu-Cu Hollow → **Dissociated** & Slipped to Cu Top (-2.498 eV).
- **Iteration 3:** End-on @ Cu Top → **Slipped** to Cu-Cu-Cu Hollow (-1.547 eV).
- **Iteration 4:** Side-on @ Ag-Cu Bridge → **Slipped** to Cu-Cu-Cu Hollow (-1.430 eV).
- **Iteration 5:** Side-on @ Cu-Cu-Cu Hollow → **Stable** (-1.740 eV).
- **Conclusion:** The exploration identifies a **metastable molecular adsorption** state at the pure Cu-Cu-Cu Hollow site (-1.740 eV). However, the global minimum is a **dissociated state** (-2.498 eV), indicating a thermodynamic preference for decomposition. The system exhibits strong "Chemical Slip," where attempts to bind to Ag-containing sites or low-

coordination Cu sites (Bridge/Top) consistently fail, funneling the adsorbate either into the pure Cu hollow (molecular) or leading to spontaneous dissociation.

No. 7 ([H] on $Ru_3Mo(111)$)

- **Iteration 1:** End-on @ Ru-Ru-Ru Hollow → **Slipped** to Ru-Ru Bridge (-3.840 eV).
- **Iteration 2:** End-on @ Mo-Ru-Ru Hollow → **Slipped** to Ru Top (-3.816 eV).
- **Iteration 3:** End-on @ Mo-Ru Bridge → **Slipped** to Ru Top (-3.775 eV).
- **Iteration 4:** End-on @ Ru-Ru Bridge → **Slipped** to Ru Top (-3.816 eV).
- **Iteration 5:** End-on @ Ru Top → **Stable** (-3.816 eV).
- **Conclusion:** The exploration identifies the **Ru-Ru Bridge** site (-3.840 eV) as the most stable configuration found, though it is energetically nearly degenerate with the **Ru Top** site (-3.816 eV, $\Delta E \approx 0.024$ eV). The system exhibits significant "**Chemical Slip**", where the adsorbate consistently rejects Mo-containing sites and high-coordination hollow sites, spontaneously migrating to lower-coordination **pure Ru** sites (Bridge or Top). No dissociation was observed (as the adsorbate is atomic H), but the strong preference for specific local environments highlights the surface heterogeneity.

No. 8 ([N] = N[H] on $Ru_3Mo(111)$)

- **Iteration 1:** Side-on @ Ru-Ru Bridge → **Slipped (Chemical)** to Mo-Mo-Ru-Ru-Ru Hollow (Isomerized, -4.821 eV).
- **Iteration 2:** Side-on @ Mo-Ru Bridge → **Slipped** to Mo-Mo-Ru Bridge/Hollow (-3.824 eV).
- **Iteration 3:** Side-on @ Mo-Mo-Ru Hollow → **Dissociated** to Mo-Mo-Ru-Ru-Ru-Ru (-3.550 eV).
- **Iteration 4:** Side-on @ Mo-Mo-Ru-Ru-Ru Hollow → **Stable** (Local Min) at Mo-Mo-Ru-Ru-Ru Hollow (-3.009 eV).
- **Iteration 5:** Side-on @ Mo-Mo-Ru-Ru-Ru Hollow → **Stable** (Deep Sink) at Mo-Mo-Ru-Ru-Ru Hollow (-4.677 eV).
- **Conclusion:** The exploration reveals a distinct "Chemical Slip" phenomenon where low-coordination bridge sites (Ru-Ru, Mo-Ru) are kinetically unstable, driving the adsorbate into a deep, high-coordination thermodynamic sink involving a mixed **Mo-Ru hollow site** (specifically 5-fold). Unlike previous systems where dissociation often represents the global minimum, here the **isomerized molecular state** ($E_{ads} \approx -4.821$ eV) is significantly more stable than the dissociated state ($E \approx -3.550$ eV), indicating a strong thermodynamic preference for preserving the N-N bond within this specific multi-metallic environment.

No. 9 ([OH] on $Pt(111)$)

- **Iteration 1:** End-on @ Pt Ontop → **Stable** at Pt Ontop (-1.823 eV).
- **Iteration 2:** End-on @ Pt-Pt Bridge → **Slipped (Chemical)** to Pt Ontop (Isomerized, -1.866 eV).

- **Iteration 3:** End-on @ Pt-Pt-Pt Hollow → **Slipped (Chemical)** to Pt-Pt Bridge (Isomerized, -1.932 eV).
- **Iteration 4:** End-on @ Pt-Pt-Pt-Pt Hollow → **Slipped (Chemical)** to Pt-Pt Bridge (-1.806 eV).
- **Conclusion:** The exploration reveals a complex potential energy surface where the **Pt-Pt Bridge site** represents the thermodynamic global minimum ($E_{ads} \approx -1.932$ eV). A notable "Chemical Slip" was observed: initializing at the bridge site kinetically trapped the adsorbate into a local Ontop minimum ($E \approx -1.866$ eV), whereas initializing at the 3-fold hollow site allowed the system to relax into the deeper Bridge sink. The study confirms [OH] prefers a high-coordination bridge motif over the conventional ontop site on Pt(111), with no dissociation observed.

No. 10 ([OH] on Pt(100))

- **Iteration 1:** End-on @ Pt Top → **Slipped (Chemical)** to Pt-Pt Bridge (-2.610 eV).
- **Iteration 2:** End-on @ Pt-Pt-Pt-Pt Hollow → **Slipped (Chemical)** to Pt-Pt Bridge (-2.609 eV).
- **Iteration 3:** End-on @ Pt-Pt Bridge → **Stable** at Pt-Pt Bridge (-2.610 eV).
- **Iteration 4:** End-on @ Pt-Pt-Pt Hollow → **Slipped (Chemical)** to Pt-Pt Bridge (-2.611 eV).
- **Conclusion:** The exploration demonstrates a decisive thermodynamic preference for the **Pt-Pt Bridge site** as the exclusive stable configuration for the hydroxyl radical on Pt(100). All attempts to stabilize the adsorbate at Top or Hollow (3-fold/4-fold) sites resulted in spontaneous **Chemical Slip** toward the bridge position. The resulting configurations are energetically degenerate ($\Delta E < 0.002$ eV), indicating that while multiple specific bridge sites exist, they represent a single, flat global minimum on the potential energy surface where the molecule remains intact without dissociation.

No. 11 ([OH] on Pd(111))

- **Iteration 1:** End-on @ Pd Top → **Slipped (Chemical)** to Pd-Pd Bridge (-2.397 eV).
- **Iteration 2:** End-on @ Pd-Pd-Pd Hollow → **Slipped (Chemical)** to Pd-Pd Bridge (-2.421 eV).
- **Iteration 3:** End-on @ Pd-Pd Bridge → **Slipped (Chemical)** to Pd-Pd-Pd Hollow (-2.263 eV).
- **Conclusion:** The exploration identifies the **Pd-Pd Bridge site** as the thermodynamically preferred configuration for the hydroxyl radical on Pd(111), with the lowest energy found at -2.421 eV. Initial attempts at Top and Hollow sites proved unstable, resulting in spontaneous **Chemical Slip** toward the Bridge position. The stable Bridge configurations found in Iterations 1 and 2 are energetically degenerate ($\Delta E \approx 0.024$ eV), indicating competitive adsorption sites at room temperature. A counter-intuitive slip was observed in Iteration 3 (Bridge → Hollow), suggesting a complex potential energy surface, though the resulting Hollow state was less stable than the Bridge global minimum. The molecule remained intact without dissociation in all iterations.

No. 12 ([OH] on Au(111))

- **Iteration 1:** End-on @ Au Top → **Slipped (Geometric)** to Desorbed State (-2.018 eV).
- **Iteration 2:** End-on @ Au-Au Bridge → **Slipped (Geometric)** to Desorbed State (-2.059 eV).
- **Iteration 3:** End-on @ Au-Au-Au Hollow → **Slipped (Geometric)** to Desorbed State (-2.021 eV).
- **Conclusion:** The exploration characterizes the interaction of the hydroxyl radical on Au(111) as a **Desorbed State** (based on strict geometric criteria used in this run), with the lowest energy found at -2.059 eV. Initial attempts at Top, Bridge, and Hollow sites resulted in spontaneous **Geometric Slip** toward a non-covalently bound state (O-Au distance ≈ 2.27 Å). The resulting configurations across all iterations are energetically degenerate ($\Delta E \approx 0.041$ eV), indicating a lack of strong site preference in this binding mode. The molecule remained intact without dissociation in all iterations.

基于你提供的运行日志 ([OH] on Ag(100)) , 填写的模板如下:

No. 13 ([OH] on Ag(100))

- **Iteration 1:** End-on @ Ag-Ag Bridge → **Slipped (Chemical)** to 4-Fold Hollow State (-2.847 eV).
- **Iteration 2:** End-on @ Ag Top → **Slipped (Chemical)** to 4-Fold Hollow State (-2.859 eV).
- **Iteration 3:** End-on @ Ag-Ag-Ag-Ag Hollow → **Stable** at 4-Fold Hollow State (-2.854 eV).
- **Conclusion:** The exploration characterizes the interaction of the hydroxyl radical on Ag(100) as a stable **Chemisorbed State at the 4-fold Hollow site**, with the lowest energy found at -2.859 eV. Initial attempts at Top and Bridge sites resulted in spontaneous **Chemical Slip** toward the thermodynamically preferred hollow site. The resulting configurations across all iterations are energetically degenerate ($\Delta E \approx 0.012$ eV), confirming convergence to the same global minimum. The molecule remained intact without dissociation in all iterations.

No. 14 ([OH] on CoPt(111))

- **Iteration 1:** End-on @ Pt Top → **Stable** at Pt Top State (-2.608 eV).
- **Iteration 2:** End-on @ Co Top → **Slipped (Chemical)** to Co-Pt Bridge State (-2.309 eV).
- **Iteration 3:** End-on @ Pt-Pt Bridge → **Slipped (Chemical)** to Pt Top State (-2.606 eV).
- **Iteration 4:** End-on @ Co-Pt-Pt Hollow → **Slipped (Chemical)** to Pt Top State (-2.605 eV).
- **Conclusion:** The exploration characterizes the interaction of the hydroxyl radical on CoPt(111) as a stable **Chemisorbed State at the Pt Top site**, with the lowest energy found at -2.608 eV. Initial attempts at Pt-Pt Bridge and Co-Pt-Pt Hollow sites resulted in spontaneous **Chemical Slip** toward the thermodynamically preferred Pt Top site, while

the Co Top site slipped to a less stable Co-Pt Bridge state. The resulting configurations across the Pt Top, Pt-Pt Bridge, and Hollow iterations are energetically degenerate ($\Delta E \approx 0.003$ eV), confirming convergence to the same global minimum. The molecule remained intact without dissociation in all iterations.

No. 15 ($[CH_2CH_2OH]$ on $Cu_6Ga_2(100)$)

- **Iteration 1:** End-on @ Cu Top → **Stable** at Cu Top State (-3.117 eV).
- **Iteration 2:** End-on @ Cu-Cu Bridge → **Stable** at Cu-Cu Bridge State (-2.463 eV).
- **Iteration 3:** End-on @ Cu-Cu-Cu-Cu Hollow → **Slipped (Chemical)** to Cu-Cu Bridge State (-3.528 eV).
- **Iteration 4:** End-on @ Cu-Ga Bridge → **Skipped** (Site not found on surface).
- **Conclusion:** The exploration characterizes the interaction of the 2-hydroxyethyl radical on $Cu_6Ga_2(100)$ as a stable **Chemisorbed State at a Cu-Cu Bridge site**, with the global minimum found at -3.528 eV. Interestingly, the direct attempt at a Cu-Cu Bridge site (Iteration 2) yielded a significantly less stable state (-2.463 eV), whereas relaxation from the Hollow site (Iteration 3) slipped into a highly stable Bridge configuration (likely involving optimized geometry or specific alloy environment). The Top site remains a stable local minimum (-3.117 eV). The molecule remained intact without dissociation in all successful iterations.

No. 16 ($[CH_2CH_2OH]$ on $Au_2Hf(102)$)

- **Iteration 1:** End-on @ Hf Top → **Slipped (Chemical)** to Au-Hf Bridge State (-3.448 eV).
- **Iteration 2:** End-on @ Au Top → **Slipped (Chemical)** to Au-Hf₃ Hollow State (-3.304 eV).
- **Iteration 3:** End-on @ Au-Au-Hf Hollow → **Dissociated** into Hf-Au-Au Hollow State (-5.135 eV).
- **Iteration 4:** End-on @ Au-Au-Au Hollow → **Stable** at Au₃ Hollow State (-1.736 eV).
- **Iteration 5:** End-on @ Au-Au Bridge → **Slipped (Chemical)** to Au₃ Hollow State (-2.202 eV).
- **Conclusion:** The exploration characterizes the interaction of the 2-hydroxyethyl radical on $Au_2Hf(102)$ as thermodynamically unstable. While the best **Molecular State** was found at a **Au-Hf Bridge site** (-3.448 eV), the system exhibited a strong tendency towards spontaneous **Dissociation**, finding a decomposed state at -5.135 eV (1.69 eV more stable than the molecular form). Significant **Chemical Slip** was observed in almost all iterations, with the radical consistently migrating from low-coordination Top sites to high-coordination Bridge or Hollow sites, often accompanied by intramolecular rearrangement ($\Delta Bond > 0$).

No. 17 (CH_3CHO on $Rh_2Ti_2(111)$)

- **Iteration 1:** End-on (O) @ Rh Top → **Slipped (Chemical)** to Rh-Rh-Ti Hollow State (-2.622

- **Iteration 2:** End-on (O) @ Ti Top → **Slipped (Chemical)** to Rh-Ti Bridge State (-2.647 eV).
- **Iteration 3:** Side-on (C=O) @ Rh-Rh-Rh-Ti Hollow → **Dissociated** into Rh-Rh-Ti-Ti Hollow State (-2.513 eV).
- **Iteration 4:** Side-on (C=O) @ Rh-Ti Bridge → **Dissociated** into Rh₅Ti₂ Hollow State (-2.913 eV).
- **Conclusion:** The exploration characterizes the interaction of acetaldehyde on Rh₂Ti₂ (111) as thermodynamically unstable. While the best **Molecular State** was found at a **Rh-Ti Bridge site** (-2.647 eV), the system exhibited a strong tendency towards spontaneous **Dissociation**, finding a decomposed state at -2.913 eV (0.27 eV more stable than the molecular form). Significant **Chemical Slip** was observed in end-on iterations, where the molecule spontaneously migrated from initial Top sites to higher-coordination Bridge or Hollow sites, driven by the strong affinity of the oxygen atom for surface Titanium.

No. 18 (*CH₃CHO* on *Al₃Zr*(101))

- **Iteration 1:** Side-on (C=O) @ Al-Zr Bridge → **Slipped (Chemical)** to Al-Zr Hollow State (-2.508 eV).
- **Iteration 2:** Side-on (C=O) @ Al-Al-Zr Hollow → **Slipped (Chemical)** to Al-Zr Hollow State (-2.752 eV).
- **Iteration 3:** Side-on (C=O) @ Al-Al Bridge → **Dissociated** into Al-Zr Hollow State (-2.881 eV).
- **Iteration 4:** End-on (O) @ Zr Top → **Slipped (Chemical)** to Al-Al-Zr Hollow State (-2.640 eV).
- **Iteration 5:** End-on (O) @ Al Top → Stable Al Top State (-2.080 eV).
- **Conclusion:** The exploration characterizes the interaction of acetaldehyde on *Al₃Zr*(101) as thermodynamically unstable regarding molecular retention. While the best **Molecular State** was found at a **High-coordination Hollow site** (slipped from Al-Al-Zr), with an energy of -2.752 eV, the system exhibited a strong tendency towards spontaneous **Dissociation**, finding a decomposed state at -2.881 eV (0.129 eV more stable than the best molecular form). Significant **Chemical Slip** was observed in nearly all iterations (except the weak Al Top binding), where the molecule spontaneously migrated from initial Bridge or Top sites to higher-coordination Hollow sites, driven by the strong affinity of the oxygen atom for surface Aluminum and Zirconium atoms.

No. 19 (*CH₃CHO* on *Hf₂Zn₆*(110))

- **Iteration 1:** End-on (O) @ Hf Top → **Slipped (Chemical)** to Hf-Hf-Hf Hollow (HCP/Subsurf) (**-3.435 eV**).
- **Iteration 2:** End-on (O) @ Zn Top → **Stable Zn Top State** (**-1.428 eV**).
- **Iteration 3:** Side-on (C=O) @ Hf-Zn Bridge → **Dissociated** into Hf Hollow (FCC/No-Subsurf) (**-3.977 eV**).

- **Iteration 4:** Side-on (C=O) @ Zn–Zn Bridge → **Slipped (Chemical) + Dissociated** to Zn Top–like State (**-1.831 eV**).
- **Iteration 5:** End-on (O) @ Hf–Hf–Zn–Zn Hollow-4 → **Slipped (Chemical)** to Hf–Hf–Hf Hollow (HCP/Subsurf) (**-3.388 eV**).
- **Conclusion:** The exploration reveals that acetaldehyde interacts strongly with **Hf-dominated hollow sites** on $Hf_2Zn_6(110)$. The best **Molecular State** occurs at an **Hf₃ Hollow site** (slipped from Hf-top) with **-3.435 eV**, but the system shows a clear thermodynamic drive toward **Dissociation**, locating a decomposed state at **-3.977 eV** (0.542 eV more stable than the best molecular adsorption). Widespread **Chemical Slip** is observed (except for the weak Zn-top binding), where the adsorbate spontaneously migrates from Top/Bridge positions into **Hf-centered Hollow basins**, highlighting the strong affinity of the carbonyl oxygen for Hf sites.

****No. 20 (*NON(CH3)* on *Bi2Ti6(211)*):** My Mac died during execution.